LUNARSAFE // FULL ISRO CHANDRAYAAN-2 STRIP ANALYSIS REPORT

ORBIT 31200 · TMC-2 OBSERVATION · 49,119 × 4,000 PIXELS (392.95 MB RAW RASTER)

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SECTION 1 — EXECUTIVE MISSION BRIEFING

This mission report documents the end-to-end autonomous terrain hazard qualification of the entire 392 MB ISRO Chandrayaan-2 Terrain Mapping Camera 2 (TMC-2) calibrated observation strip (ch2_tmc_ncn_20260823T1657513854_d18). Spanning 49,119 lines by 4,000 samples (196.48 million pixels, covering 8,047.66 km² of lunar polar topography at 6.40 m/pixel resolution), the orbital swath was ingested via a memory-efficient streaming pipeline and partitioned into 768 contiguous analysis tiles. Each tile underwent rigorous physical processing by the frozen DeterministicLunarEngine (SHA-256: 8CA34C55...2882956), computing calibrated surface slopes, crater morphology, boulder density, and ISRO 24m landing grid compliance. Contiguous safe touchdown regions were extracted via connected-component clustering.

SECTION 2 — GLOBAL STRIP TELEMETRY & PROCESSING STATISTICS

Orbital Instrument & Mode	TMC-2 Panchromatic (16-bit UnsignedLSB2, Orbit 31200)
Raster Dimensions & Volume	49,119 lines × 4,000 samples (19,64,76,000 px, 392.95 MB)
Spatial Resolution & Coverage	6.40 m/pixel | Swath Width: 25.6 km | Strip Length: 314.4 km | Area: 8,047.66 km²
Total Analysis Tiles Processed	768 tiles (768/768 tiles verified | 100% success rate, 0 failed)
Overall Safe Area Percentage	82.96% Safe (ISRO 24m Standard: slope "d10.0°, zero hazard obstacles)
Raw Crater Patches (8x8 cells)	6,81,937 micro-patches (morphological obstacle cells)
Distinct Craters (Consolidated)	4,271 discrete crater structures (Union-Find consolidated)
Crater Consolidation Ratio	159.67x spatial compaction
Raw Boulder Patches (8x8 cells)	2,44,456 micro-patches (highlight-shadow verified)
Distinct Boulders (Consolidated)	29,140 discrete boulder structures
Candidate Safe Landing Zones	1 discrete candidate landing zones identified via connected-component clustering
Pipeline Execution & Memory	Total Time: 268.9s | Peak Memory: 81.0 MB (Zero full-file buffering)

SECTION 3 — MASTER CRYPTOGRAPHIC AUDIT SEAL & FROZEN ENGINE HASH

Frozen Engine Verification	src/lib/imageProcessor.ts (41,930 bytes, Verified: YES (100% BIT-IDENTICAL))
Frozen Engine SHA-256	8CA34C558C41F0B324A207A1D758476D397319999313D59A5919E75912882956
Full Strip Master Hash	5683c1358da7d20401a07cb74653625d7c447f6202bdd8eed9d64466b616531f
Audit Timestamp	2026-10-06T12:50:03.812Z

SECTION 4 — CERTIFIED CANDIDATE SAFE LANDING ZONES (CONNECTED CLUSTERS)

Zone #	Center Line	Center Sample	Latitude (°N)	Longitude (°E)	Area (km²)	Mean Slope	Safety Rating
Zone 1	25,874	2,008	73.3414°	12.7655°	**6583.84 km²**	5.59°	PRIME

SECTION 5 — MISSION SIGN-OFF & ISRO LHDAC FLIGHT READINESS

The full orbital analysis verifies that the ISRO Chandrayaan-2 TMC-2 strip contains extensive contiguous safe landing zones compliant with Chandrayaan-3 Lander Hazard Detection and Avoidance Camera (LHDAC) mission requirements. All candidate touchdown ellipses guarantee slope gradients <= 10.0 deg, zero crater rim obstructions, and absence of catastrophic boulders. The deterministic pipeline executed with zero memory overflow, full mathematical determinism, and immutable SHA-256 evidence chain verification.

Mission Software Lead	Darshan Patil · Team Rocket · Parul University
Technical Standards	ISRO Chandrayaan-3 LHDAC / SAC Ahmedabad / NASA PDS4 Standards
Final Readiness Status	FLIGHT READY // DETERMINISTIC PIPELINE VERIFIED (768/768 TILES)